State minimum wage laws and state-level rates of new HIV diagnoses among Black, Latine, and White US women and men, 2010–2019

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ABSTRACT

Background In the USA, HIV remains a pressing public health challenge (~37 400 new diagnoses in 2018) and pronounced racialised inequities persist. In 2018, Black and Latine people represented 43% and 26% of new diagnoses but only 13% and 18% of the population, respectively. Antiracist laws may help mitigate these inequities, yet research investigating specific laws is lacking. This study aimed to examine associations between state minimum wage laws and new HIV diagnosis rates by racialised group and sex/gender.

Methods We linked data on inflation-adjusted state minimum wage and state-level new HIV diagnosis rates among US adults aged 13–59 years. We fit two-way fixed-effects models to examine associations between changes in minimum wage and changes in new HIV diagnosis rates per 100 000 people from 2010 to 2019 by racialised group (Black/Latine/White) and sex/gender (women/men). Additional analyses tested for effect heterogeneity by racialised group and by sex/gender within racialised group.

Results Findings show an inverse association between state minimum wage and new HIV diagnosis rates across all racialised and sex/gender groups, except White women. Among Black adults, a 1 US Dollar ($) increase in state minimum wage was associated with a 5.81 per 100 000 decrease in new HIV diagnosis rate, followed by Latine (2.81/100 000) and White adults (1.50/100 000). The inverse association was larger among Latine (4.73/100 000) and White men (2.88/100 000) than Latine (0.98/100 000) and White women (0.07/100 000), respectively. There was no evidence of effect heterogeneity comparing Black men to Black women.

Conclusions State minimum wage laws may represent a critical policy intervention to address racialised HIV inequities.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Prior research suggests antiracist laws/policies—such as minimum wage laws—can help mitigate racialised health inequities, yet limited research has examined the impact of specific laws/policies by racialised group.

WHAT THIS STUDY ADDS

⇒ This study estimates associations between changes in state minimum wage and changes in state-level rates of new HIV diagnoses among US adults by racialised group and sex/gender from 2010 to 2019.

⇒ Findings suggest an inverse association between state minimum wage and rates of new HIV diagnoses across all racialised and sex/gender groups, except White women.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Minimum wage laws represent an important policy strategy to help mitigate racialised HIV inequities.

⇒ Future research is needed to identify mechanisms and potential differences in mechanisms by racialised group and sex/gender.

INTRODUCTION

HIV remains a major public health challenge in the USA, with approximately 37 400 new diagnoses in 2018.[1] HIV inequities are pronounced and persistent, with marked inequities documented with respect to age, geographic region, racialised group, sex/gender, and sexual orientation.[1] In 2018, Black people represented 43% and Latine people represented 26% of new HIV diagnoses but only 13% and 18% of the US population, respectively.[2] Furthermore, in 2018, Black women represented 26% of new HIV diagnoses among Black people, while Latine and White women represented 12% and 16% of new HIV diagnoses among Latine and White people, respectively.[2]

Structural determinants—including structural racism—are hypothesised to be fundamental causes of racialised inequities in HIV.[3] Structural racism refers to the ways in which social systems and institutions systematically restrict access to power, resources, and opportunities and exacerbate exposure to risks and harms for marginalised racialised groups while advantaging White people.[4] Structural racism—including its manifestation in racist laws/policies—shapes the inequitable distribution of social determinants of health across racialised groups, resulting in population-level racialised inequities in health outcomes, including HIV.[4] Antiracist laws/policies (ie, those that extend benefits or resources and/or prevent penalty or harm to marginalised racialised groups)[5 6] can address the inequitable distribution of these social determinants and thus help mitigate racialised health inequities, yet research on the impact of specific laws is lacking.[7]

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Minimum wage laws, which enable states and cities to set the minimum wage for their jurisdiction at a higher rate than the federal minimum wage, are one such antiracist law.[8] Importantly, the first federal minimum wage law—the Fair Labor Standards Act (FLSA), enacted in 1938 during the New Deal era—was racist in both its design and consequences, as it intentionally excluded sectors with predominantly Black workforces.[8] These gaps in wage protections persisted until the 1966 FLSA amendments extended coverage to previously excluded sectors—including agriculture, restaurants, schools, nursing homes and other service sectors, which collectively employed about one-third of all Black workers.[9] Prior research indicates that these amendments accounted for nearly 20% of the reduction in the Black-White income gap observed during the late 1960s and early 1970s.[9]

Since 2009, the federal minimum wage has remained stagnant at 7.25 US Dollars ($) per hour, without adjustment for inflation, thereby hindering its potential to address marked, long-standing racialised income and wealth inequities in the USA.[10] Increasingly, state minimum wage laws have been proposed as a policy intervention to address these racialised income and wealth inequities that in turn drive health inequities.[9 11] According to US Census Bureau data, in 2020, the median income for Black households was approximately $46 000 compared with $75 000 for White households—only about 61 cents for every dollar of median household income.[12] Since racialised income inequities are cumulative across the life-course and generations, they contribute to widening racialised wealth inequities. As a consequence of historical and contemporary racist laws/policies and practices, Black and Latine people disproportionately work minimum wage-paying jobs (eg, food service, home and childcare) and live below the federal poverty level, compared with White people.[13] Accordingly, recent analyses from the Economic Policy Institute project that raising the federal minimum wage to $17 per hour would disproportionately benefit Black and Latine workers, thereby helping to reduce racialised income gaps.[10] Thus, the economic benefits (eg, reductions in poverty and income inequality) of increasing minimum wages may mitigate many of the structural harms that impact health outcomes, including HIV, particularly among marginalised racialised groups.

With respect to HIV specifically, prior research suggests several mechanisms via which increased minimum wages may affect rates of new HIV diagnoses (henceforth, HIV diagnosis rates) and points to how these impacts may differ across sociodemographic groups.[1 3 14–16] Specifically, increased minimum wages likely result in reductions in poverty and economic instability, which in turn may facilitate access to healthcare (including HIV prevention services), food security, and housing stability—all of which are protective factors against HIV acquisition.[3] Increased minimum wages and resulting improved economic security may also lead to decreased engagement in survival strategies (eg, transactional sex) and decreased psychological stress and related coping strategies (eg, substance use), which are more prevalent among economically marginalised populations.[15] These mechanisms may be particularly salient for Black and Latine people who are disproportionately affected by the structural factors (including racist laws) driving both economic instability and patterns of HIV risk, and who will thus likely benefit most from increased minimum wages.[1 3] Importantly, an intersectional framework[17] illuminates how intersecting systems of oppression—including racism and sexism—jointly shape the lived experiences and health of Black and Latine women. For example, Black and Latine women are disproportionately represented in minimum wage jobs (eg, home and childcare, and other service sectors) as a result of the compounding impacts of structural racism and sexism, which produce racialised and gendered inequities; thus, increased minimum wages may be uniquely protective against HIV for Black and Latine women.[18]

The number of US states that have set their minimum wage at a higher rate than the federal minimum wage has grown over time. The most recent change to the federal minimum wage occurred on 24 July 2009, when the minimum wage was raised from $6.55 to $7.25 per hour.[8] In 2010, 15 US states had minimum wages above the federal minimum wage, and by 2019, 30 did. This patterning of increasing state minimum wages across US states—most of which are concentrated in the West and Northeast—may have mitigated racialised HIV inequities, since increasing minimum wages can in turn reduce poverty and income inequality—which are documented drivers of these inequities[19 20]—yet this remains understudied.[21] To address this gap, this study examined associations between changes in state minimum wage laws and changes in state-level HIV diagnosis rates among US adults aged 13–59 years by racialised group and—based on prior research documenting heterogeneity by sex/gender[22–24]—examined further differences by sex/gender, in 50 US states and D.C. from 2010 to 2019.

METHODS

Measures

Exposure

We operationalised state minimum wage as the difference between the state and federal minimum wage for each state by year from 2010 to 2019.[25] We adjusted for inflation by expressing all differences in minimum wage in terms of 2020 dollars.[25] For additional details on the state minimum wage laws dataset, see online supplemental note 1.

Outcome

The outcomes were racialised group-specific (Black/Latine/White) and sex/gender-specific (women/men) HIV diagnosis rates for each state by year from 2010 to 2019 obtained from AIDSVu.[26] Notably, AIDSVu defines sex as 'sex-at-birth'. We use the term 'sex/gender' throughout to refer to this variable, since AIDSVu does not state how sex was assessed, thus precluding us from determining the extent to which there may be measurement error in this variable due to conflation of sex-assigned-at-birth and gender identity.[26] While we acknowledge the conflation of sex-assigned-at-birth and gender identity in this dataset, we interpret differences in HIV diagnosis rates as being largely driven by social and structural factors—such as differential exposure to economic marginalisation, sexuality and gender norms, and violence and discrimination—rather than by biological differences.[27] As such, our analyses and interpretations integrate a gender-based perspective, in accordance with established best practices in sex-based and gender-based analysis in health research.[28]

Time-varying covariates

Guided by a directed acyclic graph (online supplemental figure 1), we selected a set of time-varying covariates hypothesised to function as confounders, or common causes of state minimum wage laws and state-level HIV diagnosis rates: voter political lean and political ideology of federal elected officials (conceptualised as determinants of the state policy context that influence which laws are enacted or repealed and how public resources are allocated, including with respect to investments in HIV-related

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resources)[29] and Medicaid expansion (conceptualised as a co-occurring antiracist law that was enacted across multiple states during the study period and has had substantial impacts on increasing healthcare access, particularly among marginalised racialised groups).[30 31] We conceptualised racialised inequity indicators (eg, Black/White ratio measures of social, political, and economic indicators) as downstream of (anti-)racist laws and therefore did not adjust for them in our analyses.[5 11]

Data on political ideology of the House of Representatives and of the Senate were obtained from Voteview.[29] We obtained political ideology scores for each state based on federal elected officials' roll call votes for each Congress from the 110th Congress (2007–2009) through the 115th Congress (2017–2019); values range between −1 (most liberal) and 1 (most conservative).[29] Since D.C. lacks congressional representation, we assigned D.C. to take on the value of the most liberal US state for each of these variables because (1) D.C. had the most liberal voter political lean for 2010–2019 and (2) D.C.'s policy environment is considered one of the most liberal in the USA.[5]

Data on voter political lean for the 2008, 2012, and 2016 US presidential elections were obtained from the Cook Political Report; these scores quantify the extent to which the vote for a given district differed from the US average score.[29]

Data on Medicaid expansion status from the Kaiser Family Foundation categorised states as having expanded Medicaid to cover all individuals with incomes up to 138% of the federal poverty level versus not having expanded Medicaid.[32]

Statistical analysis

Linear models with state and year fixed effects were fit to estimate associations between changes in state minimum wage laws and changes in state-level HIV diagnosis rates per 100 000 people by racialised group and by sex/gender within racialised group from 2010 to 2019. Models included state fixed effects to capture differences in HIV diagnosis rates between states due to unmeasured time-invariant state-level factors (including state baseline sociodemographic composition, HIV-related public health investment, and policy environment). Models also included time fixed effects to capture the secular trend in HIV diagnosis rates over all states in the years of the study period (as well as any federal-level policy changes occurring over the study period). Models adjusted for the time-varying covariates listed above. We modelled state minimum wage as a continuous exposure variable; thus, the beta coefficient has an interpretation as the average change in HIV diagnosis rates for a $1 change in the state minimum wage, whenever in the study period it occurred. We fit additional models with HIV diagnosis rates as the outcome that included interaction terms between state minimum wage and racialised group and sex/gender to formally test for effect heterogeneity by racialised group and by sex/gender within racialised group. We also conducted several sensitivity analyses (see online supplemental appendix A for details). All analyses were conducted in R V.4.2.3.

RESULTS

Figure 1 depicts the difference between state and federal minimum wage in the USA from 2010 to 2019. In 2010, 29% (n=15) states had minimum wages above the federal minimum wage of $7.25 per hour, and by 2019, 59% (n=30) did.

Figure 2 depicts US national HIV diagnosis rates per 100 000 among adults aged 13–59 years by racialised group and by sex/gender within racialised group from 2010 to 2019. These rates were highest among Black people (2010: 62/100 000; 2019: 45/100 000), followed by Latine people (27/100 000; 22/100 000), and then by White people (7/100 000; 5/100 000) from 2010 to 2019. There were also notable inequities by sex/gender within each racialised group, with men experiencing higher HIV diagnosis rates compared with women over this period (Black men (2010: 91/100 000; 2019: 72/100 000), Black women (36/100 000; 21/100 000), Latine men (44/100 000; 38/100 000), Latine women (9/100 000; 5/100 000), White men (12/100 000; 9/100 000), White women (2/100 000; 2/100 000)). Notably, for all groups defined by racialised group and sex/gender, except White women, national HIV diagnosis rates appear to have decreased from 2010 to 2019. The largest absolute decreases were experienced among Black men (19/100 000)

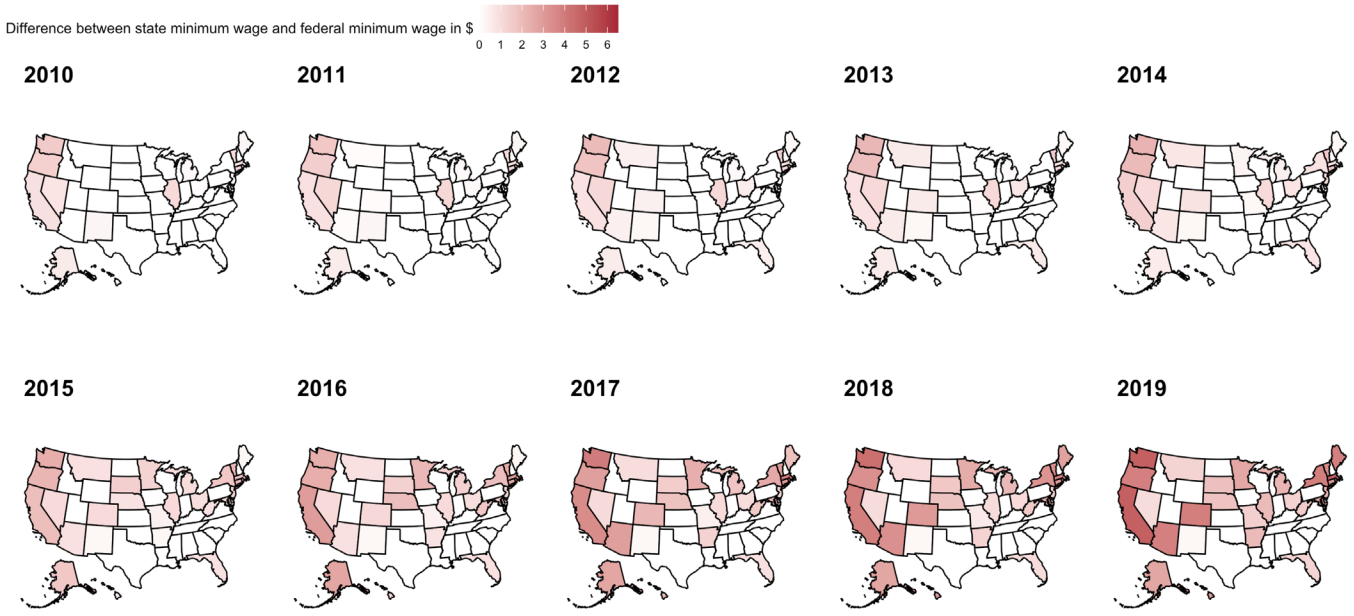

Figure 1 Difference between state minimum wage and federal minimum wage in dollars, USA, 2010–2019.

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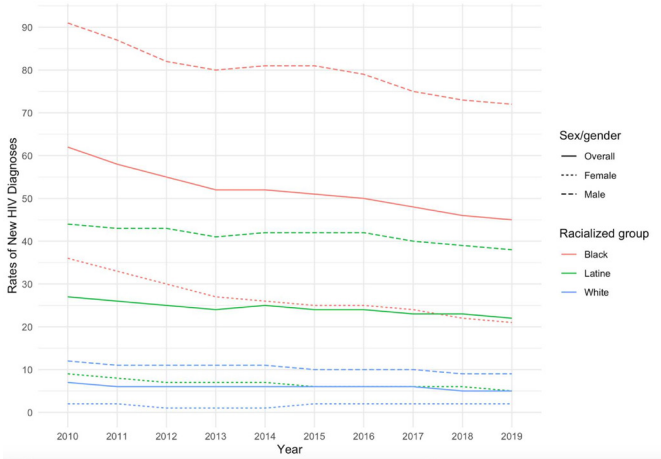

Figure 2 National rates of new HIV diagnoses (per 100 000) among adults aged 13–59 years by racialised group and sex/gender, USA, 2010–2019.

and Black women (15/100 000), while the largest proportional decreases were experienced among Latine women (44%) and Black women (42%).

We observed a consistent inverse association between state minimum wage and HIV diagnosis rates across all racialised and sex/gender groups, although CIs were sometimes wide (table 1).

We observed statistically significant effect heterogeneity by racialised group, with the magnitude of association being significantly larger among Black people and among Latine people compared with White people. On average, a $1 increase in state minimum wage was associated with a 5.98 per 100 000 (95% CI −9.34 to −2.62) decrease in the rate of new HIV diagnoses among Black people, followed by Latine people (2.88/100 000; 95% CI −4.71 to −1.05), and White people (1.55/100 000; 95% CI −2.32 to −0.78), accounting for state and time fixed effects and controlling for time-varying covariates. We also observed statistically significant effect heterogeneity by sex/gender within racialised group for Latine people and White people, with the magnitude of the association being larger for Latine men (4.86/100 000; 95% CI −8.32 to −1.41) and White men (2.98/100 000; 95% CI −4.49 to −1.47) compared with Latine women (0.98/100 000; 95% CI −1.66 to −0.30) and White women (0.08/100 000; 95% CI −0.19 to 0.04; null), respectively. We found no evidence of statistically significant effect heterogeneity comparing Black men (6.75/100 000; 95% CI −10.68 to −2.83) and Black women (6.80/100 000; 95% CI −10.74 to −2.86). Results from our sensitivity analyses were consistent with those from the main analysis (see online supplemental appendix A and tables 1–3).

DISCUSSION

Our finding of a consistent inverse association between state minimum wage and HIV diagnosis rates across racialised groups, overall and by sex/gender, is consistent with prior research that has documented inverse associations between minimum wage laws and a range of physical health outcomes.[24 25 33–35] Only one prior study has investigated the association between minimum wage laws and HIV—this study found that from 2008 to 2015, MSAs with a minimum wage that was $1 higher at baseline had a 27% lower rate of new HIV diagnoses among heterosexual Black people.[21] Our findings are consistent with this prior study, although to our knowledge, our study is the first to examine associations between minimum wage laws and HIV diagnosis rates among US adults by racialised group and by sex/gender within racialised group, which enabled us to quantify the impact of increasing minimum wages on mitigating racialised HIV inequities, particularly among men.[7]

Furthermore, we observed that the inverse association between state minimum wage and HIV diagnosis rates was significantly larger among Black people and Latine people compared with White people. Prior research has identified poverty and income inequality as determinants of HIV inequities.[19 20] Thus, this finding may be attributable to the notably higher rates of Black and Latine people employed in minimum wage-paying jobs and higher rates of poverty among Black and Latine households—which are consequences of historical and contemporary racist laws/policies and practices, including occupational, residential, and educational segregation.[4 11 13] However, relatively few studies have examined the impacts of minimum wage laws on racialised health inequities; thus, more work in this area is needed.[36]

We also found that the inverse association between state minimum wage and HIV diagnosis rates was significantly larger among Latine men and White men compared with Latine

Table 1 Two-way fixed effects models examining associations between state minimum wage and state-level rates of new HIV diagnoses (per 100 000) among adults aged 13–59 years, overall and by sex/gender and racialised group—50 US states and the District of Columbia, 2010–2019

	Model 1 β (95% CI)	Model 2 β (95% CI)	Model 1 β (95% CI)	Model 2 β (95% CI)	Model 1 β (95% CI)	Model 2 β (95% CI)
	Black overall		Latine overall		White overall	
State minimum wage*	**−5.70†** **(−8.44 to −2.95)**	**−5.81†** **(−9.15 to −2.47)**	**−2.71†** **(−4.31 to −1.10)**	**−2.81†** **(−4.60 to −1.02)**	**−1.47** **(−2.20 to −0.74)**	**−1.50** **(−2.24 to −0.76)**
	Black men		Latine men		White men	
State minimum wage	**−6.77** **(−10.28 to −3.26)**	**−6.58** **(−10.47 to −2.70)**	**−4.63‡** **(−7.66 to −1.60)**	**−4.73‡** **(−8.11 to −1.36)**	**−2.81‡** **(−4.24 to −1.38)**	**−2.88‡** **(−4.34 to −1.42)**
	Black women		Latine women		White women	
State minimum wage	**−5.54** **(−8.45 to −2.62)**	**−6.65** **(−10.62 to −2.68)**	**−0.83** **(−1.45 to −0.21)**	**−0.98** **(−1.66 to −0.30)**	**−0.12** **(−0.22 to −0.02)**	−0.07 (−0.18 to 0.04)

Bolded values indicate beta coefficients with 95% CIs that exclude the null value (ie, 0.00). The 95% CIs reflect robust standard errors.
Model 1 includes fixed effects for state and year.
Model 2 includes fixed effects for state and year and the following time-varying covariates: Medicaid expansion status, voter political lean, ideology of the House, and ideology of the Senate.
*Time-varying covariates are lagged by 1 year.
†Indicates that there is statistically significant effect heterogeneity across racialised groups, comparing a given racialised group to White people.
‡Indicates that there is statistically significant effect heterogeneity across sex/gender groups, comparing men to women, within a given racialized group.

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women and White women, respectively, which may be attributable to the existence of sex/gender-related differences due to: (1) markedly higher HIV diagnosis rates among Latine and White men compared with Latine and White women, respectively,[1] (2) notably higher rates of Latine and White women employed in minimum wage-paying jobs compared with Latine and White men, respectively,[18] and/or (3) differences in the magnitude of the effect of state minimum wage laws on HIV diagnosis rates, potentially due to differences in mechanisms.[37] In contrast, we observed no statistically significant effect heterogeneity comparing Black men to Black women. This finding may be attributable to: (1) greater equality in wages between Black men and Black women compared with men and women of other racialised groups,[38] and/or (2) similarities in the magnitude of the effect of state minimum wage laws, potentially due to overlap in mechanisms.[37] However, limited empirical research has investigated potential mediators along the causal pathway from state minimum wage laws to health inequities, with even fewer examining potential differences by racialised group and sex/gender. One prior study examining the association between MSA-level minimum wage laws and HIV among heterosexual Black people explored potential mediators, including income inequality and rates of employment, poverty, incarceration, and public benefit utilisation, and found that income inequality may mediate the association.[21] Additional research identifying potential mechanisms (including those outlined in the Introduction) and investigating differences by racialised group and sex/gender is needed.

Our findings address a gap in knowledge on how specific structural racism-related (ie, racist and anti-racist) laws/policies shape racialised inequities in HIV. Compared with the rapidly growing body of research examining impacts of structural racism, commonly measured using indices comprised of multiple laws/policies and/or multiple inequity indicators,[39 40] relatively limited research has examined the impacts of specific structural racism-related laws/policies, including historical laws/policies (eg, Jim Crow laws) and contemporary laws/policies (eg, cash bail reform) on racialised inequities.[7] Research focused on examining the impacts of specific structural racism-related laws/policies on racialised inequities has direct implications for intervention (eg, abolishing racist laws/policies, enacting antiracist laws/policies) to eliminate the root causes of these inequities.[40] That said, a single law/policy change is insufficient to address the pronounced and persistent impacts of structural racism on racialised inequities; thus, there is a need for a multifaceted, multisectoral approach to policy change and analysis.[6 41 42]

Importantly, in reality, state minimum wage laws operate within a broader policy environment comprised of a multitude of co-occurring laws that synergistically or antagonistically interact to shape population health and inequities.[6 41 42] Moreover, these policy environments tend to be characterised by clusters of interrelated laws; for example, states with minimum wages above the federal level also tend to have Earned Income Tax Credit programmes, paid sick leave laws, and Medicaid expansion, whereas those with minimum wages at the federal level do not tend to have those laws but do tend to have right-to-work laws.[42] While our models controlled for the baseline state policy environment via state fixed effects, the presence of co-occurring laws that are both unmeasured and time-varying could have contributed to residual confounding of our estimates. Notably, the growing recognition of policy co-occurrence has spurred calls for the use of novel approaches—such as latent class measures, policy indices, and mixed-methods—to more accurately capture the complexity of real-world policy environments, including how co-occurring laws interact to jointly shape population health and inequities.[5 6 40 41] Thus, our findings should be interpreted within the context of the limitations of our methodological approach. However, within the field of health policy research, both single-policy and clustered-policy approaches are critically important, as single-policy approaches such as ours can generate actionable evidence and targeted recommendations for advocates and policymakers regarding whether to enact or repeal specific policies, while clustered-policy approaches can inform more comprehensive, multipronged policy interventions to address racialised health inequities.[41]

Limitations

This study has several limitations. First, this study focused on examining inequities by racialised group and sex/gender; however, there are documented inequities in HIV diagnosis rates by age, geographic region and sexual orientation—thus, future research examining inequities by additional axes of social identity is warranted.[1] Additionally, given prior research documenting the disproportionate burdens of economic instability and HIV faced by transgender populations, there is a critical need for future research conducted using transgender-inclusive data.[15] Second, we used an ecological study design and analysed only state-level data; thus, study findings do not provide insight into the associations between state minimum wage laws and individual-level outcomes. Thus, future research that incorporates individual-level outcome data and uses a multilevel design is needed. Third, we did not account for city-level minimum wage laws, which changed over the study period. In 2010, there were only three cities with minimum wage laws that exceeded the state or federal minimum wage.[43] By 2020, there were 42.[43] While city-level minimum wage laws covered only 8% of the US workforce in 2020,[43] it is still important for future research to consider accounting for this potential source of exposure misclassification. Fourth, future research conducted using methodological approaches that are appropriate for investigating how co-occurring (anti-)racist laws jointly shape racialised inequities in HIV is warranted.[41] Additionally, future research should consider examining political ideology of federal elected officials not only as a potential confounder but also as a potential effect modifier or mediator in analyses of state minimum wage laws and racialised health inequities.

CONCLUSION

To inform the development and implementation of equity-focused laws/policies, future work is needed to (1) identify potential mechanisms and differences in mechanisms by racialised group and sex/gender, (2) investigate potential inequities across intersectional subgroups defined by additional dimensions of social identity, and (3) build on the growing body of work on state minimum wage laws by simulating the effects of not yet existing antiracist laws, such as living wage laws (ie, laws that set the living wage at the earnings necessary to cover the cost of meeting one's basic needs based on locality-specific cost of living) and compare these effects to the effects of minimum wage laws.[44] Findings from this study underscore the importance of quantifying the impacts of structural racism-related laws/policies to support efforts to abolish racist laws/policies and enact antiracist laws/policies to improve population health and mitigate racialised inequities.[5 6 11] Furthermore, findings indicate that increasing state minimum wages may be an effective policy strategy to help mitigate racialised HIV inequities.

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Patient consent for publication Not applicable.

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Provenance and peer review Not commissioned; internally peer-reviewed.

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